-done- Problem 1: Heap vs BST for K Smallest Elements

Suppose you're given an unsorted array of n elements, and asked to store all of them in either a heap or BST. Which data structure would you choose, if you wish to later efficiently find the k smallest elements (where k << n). Give the time complexity for both approaches (including storage of the n elements) and explain which is better for this specific problem.

Problem 2: DFS Cycle Detection

Write pseudocode for an algorithm that uses DFS to detect if an undirected graph contains a cycle.

-done- Problem 3: BFS Path with Exactly K Edges

Given an unweighted directed graph, you want to find if there exists a path from s to t that uses exactly k edges (not shortest path, but exactly k edges). Modify the BFS algorithm above to solve this problem.

-done- Problem 4: Comparing MST Algorithms

PART A - Given a connected, weighted graph where all edge weights are distinct, prove that there is exactly one minimum spanning tree.

PART B - If all edges in a graph have the same weight, can you predict the weight of the minimum spanning tree?

PART C - You're given a graph and told that one specific edge e must be included in the MST. Describe how you would modify either Kruskal's or Prim's algorithm to find an MST that includes e. What condition on e guarantees that such an MST exists?

Problem 5: Heap Operations for Streaming Data

You're receiving a stream of integers and need to maintain the median at all times. Design a solution which uses two heaps, write pseudocode and analyze the time complexity for:

1. Adding a number to your data structure
2. Retrieving the current median

Problem 6: Prim's with Distance Constraints

You're designing a network where each connection has a cost (edge weight) and you want to minimize total cost, but with a constraint: no node can be more than D hops (edges) away from a designated center node.

Modify Prim's algorithm to build a minimum spanning tree rooted at a center node where every node is at most D edges away from the center. Provide pseudocode for your modification. What is the time complexity?

-done- Problem 7: Dijkstra's with Limited Fuel

You're planning a route where your vehicle has limited fuel. Each edge has a distance (weight), and you start with F units of fuel. You can refuel at certain nodes (gas stations).

Modify Dijkstra's algorithm to find the shortest path from source s to target t, where you can only travel on edges if you have sufficient fuel. You must visit gas stations to refuel along the way. The graph has a set of nodes G (gas stations) where you refuel to F units. Write pseudocode for your algorithm, and compare the time complexity with the standard Dijsktra implementation.